



Assignment 1

MAT110: Differential Calculus and Coordinate Geometry

Section: 16

Semester: Spring 2026

*Please write your name, ID, course name and section on the first page of the assignment answer script. **Submissions must be made by March 28, 2026, at 11:00 p.m..***

*You can only submit a PDF file - image or doc files won't be accepted. Before submitting the PDF, please rename the PDF file in the format - **ASSIGNMENT 1_ID_SECTION**. The size of the PDF file should be less than 10MB.*

*Answer all the questions by yourself. Plagiarism will lead to an F grade in the assignment. **The total score is 70. Each question is worth 10 marks. It will be converted to 15.***

1. Find the limit using Squeeze (Sandwich) theorem:

$$\lim_{x \rightarrow -5} |x + 5| \sin \left(\frac{1}{x + 5} \right).$$

2. Determine whether $k(x)$ is continuous at $x = 3$, where

$$k(x) = \begin{cases} \frac{1}{x-3}, & x < 3 \\ x+3, & 3 \leq x \leq 5 \\ x^2 - 17, & x > 5. \end{cases}$$

3. Use L'Hospital's rule to evaluate the following limit:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{\cos 3x}{x^2} \right).$$

4. Find the derivative of the following function:

$$y = \sqrt{\frac{2x^2 - 1}{x^2 + 3}}.$$

5. Given that $f(x) = x^{\frac{2}{3}}$. Find:

- (a) the open intervals on which f is increasing,
- (b) the open intervals on which f is decreasing, and
- (c) hence find the relative extrema.

6. Find the Maclaurin series of the function $f(x) = e^{6x+3}$

7. Verify the hypothesis and conclusion of Rolle's Theorem for the function $f(x) = (x^2 - 2x)e^x$ on the interval $[0, 2]$.